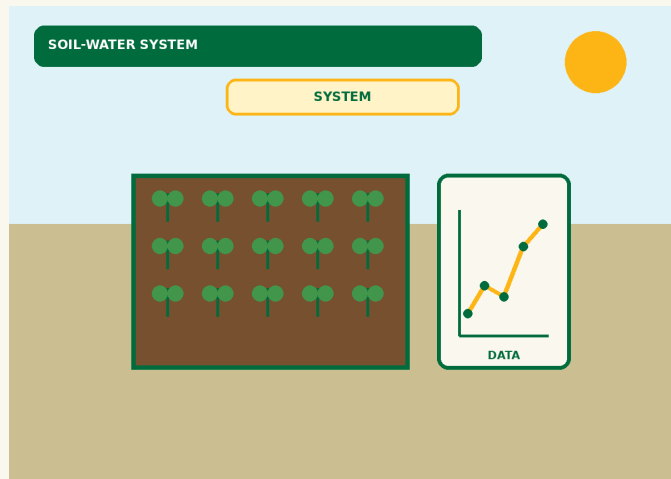




SYSTEM

SOIL-WATER SYSTEM

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 701 • SOIL-WATER SYSTEM

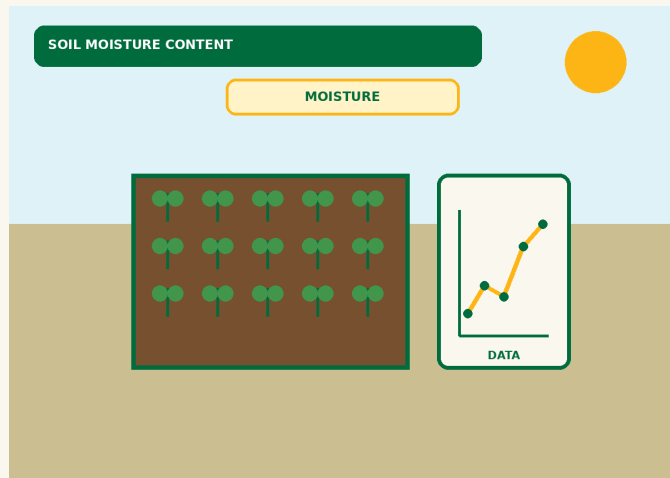
CONCEPT

Crop water management links water supply, soil storage, root uptake, atmospheric demand, irrigation, drainage, and crop response.

INVESTIGATION TASK

Draw the main water pathways.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



MOISTURE

SOIL MOISTURE CONTENT

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 702 • SOIL MOISTURE CONTENT

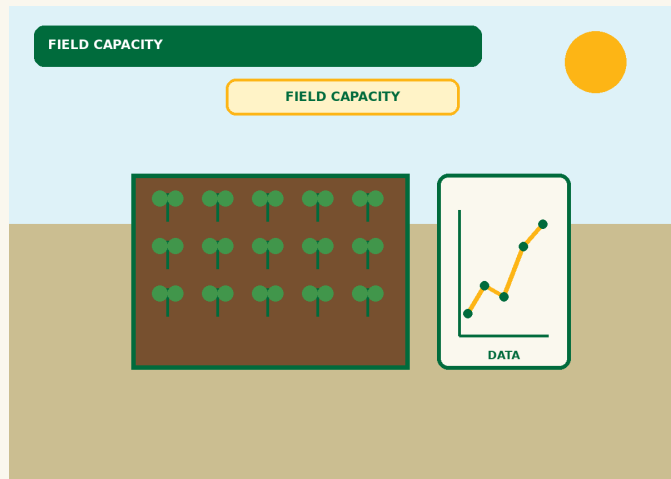
CONCEPT

Soil moisture content quantifies water held in the soil and can be measured or estimated using defined methods.

INVESTIGATION TASK

Compare readings from two depths or locations.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



FIELD CAPACITY

FIELD CAPACITY

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 703 • FIELD CAPACITY

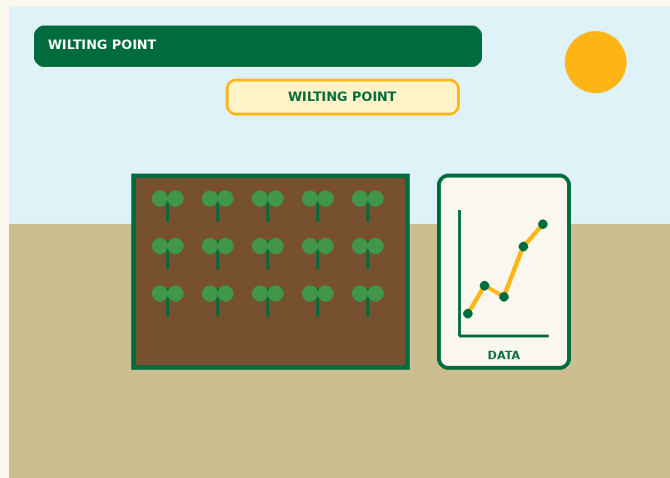
CONCEPT

Field capacity is the soil-water condition after excess gravitational water has drained and downward movement has slowed.

INVESTIGATION TASK

Interpret a teacher-provided soil-water diagram.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



WILTING POINT

WILTING POINT

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 704 • WILTING POINT

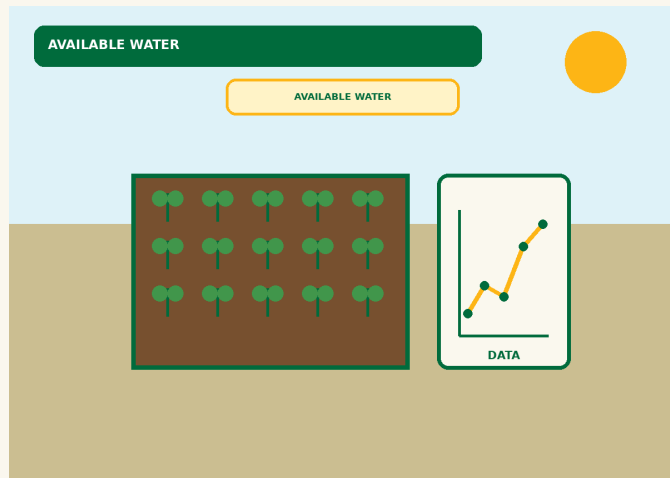
CONCEPT

Permanent wilting point is a low soil-water condition at which plants cannot recover turgor without added water.

INVESTIGATION TASK

Locate it on a simplified soil-water scale.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



AVAILABLE WATER

AVAILABLE WATER

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 705 • AVAILABLE WATER

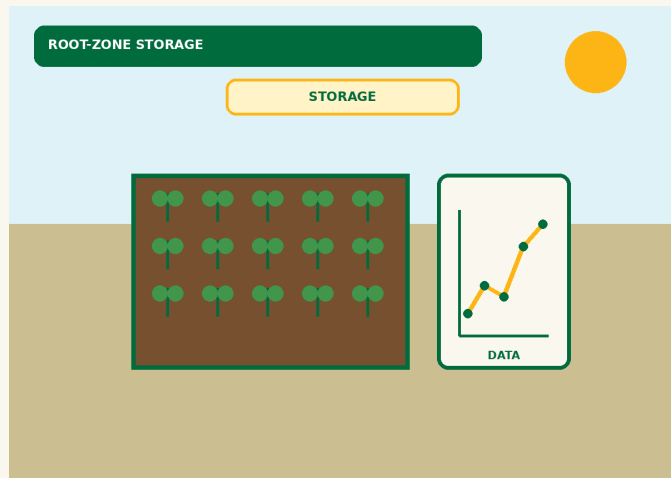
CONCEPT

Plant-available water is the portion of stored soil water between field capacity and permanent wilting point.

INVESTIGATION TASK

Calculate available water from provided values.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



STORAGE

ROOT-ZONE STORAGE

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 706 • ROOT-ZONE STORAGE

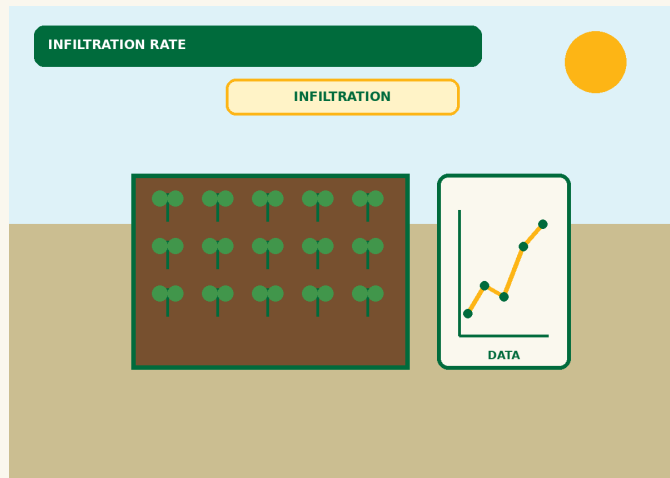
CONCEPT

Root-zone water storage depends on soil properties, water content, and effective rooting depth.

INVESTIGATION TASK

Estimate storage using simplified data.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



INFILTRATION

INFILTRATION RATE

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 707 • INFILTRATION RATE

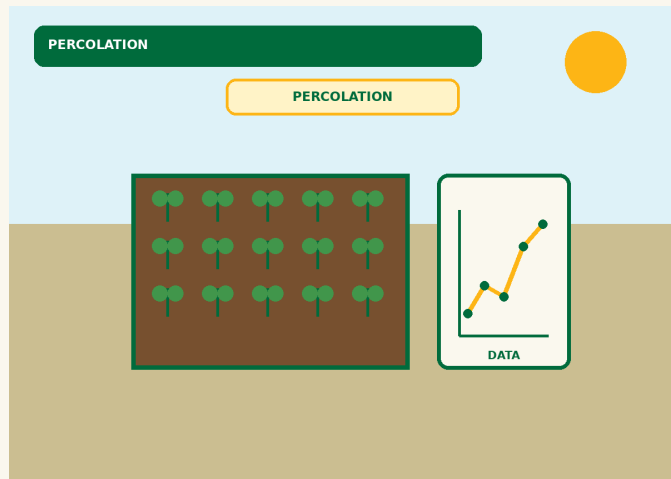
CONCEPT

Infiltration rate is the rate at which water enters the soil surface under specified conditions.

INVESTIGATION TASK

Compare measured infiltration between two surfaces.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



PERCOLATION

PERCOLATION

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 708 • PERCOLATION

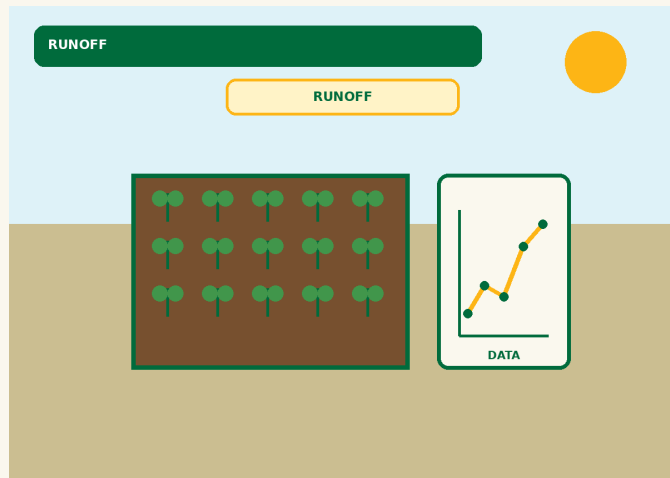
CONCEPT

Percolation is downward water movement through soil below the wetted or root zone.

INVESTIGATION TASK

Identify when excessive irrigation could increase deep drainage.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



RUNOFF

RUNOFF

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 709 • RUNOFF

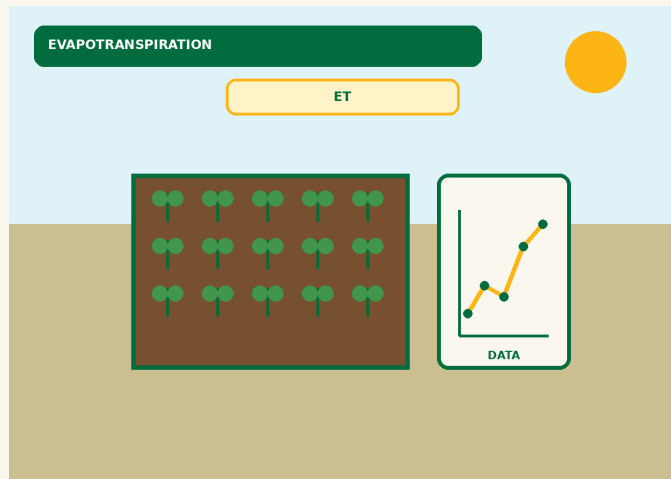
CONCEPT

Runoff is surface water movement away from the intended application area.

INVESTIGATION TASK

Measure or map where runoff occurs in a demonstration.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



ET

EVAPOTRANSPIRATION

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 710 • EVAPOTRANSPIRATION

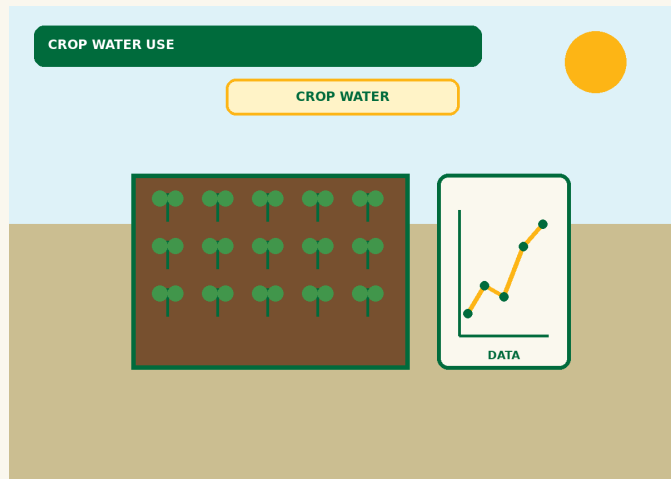
CONCEPT

Evapotranspiration combines evaporation from surfaces and transpiration from plants.

INVESTIGATION TASK

Explain the two components of ET.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



CROP WATER

CROP WATER USE

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 711 • CROP WATER USE

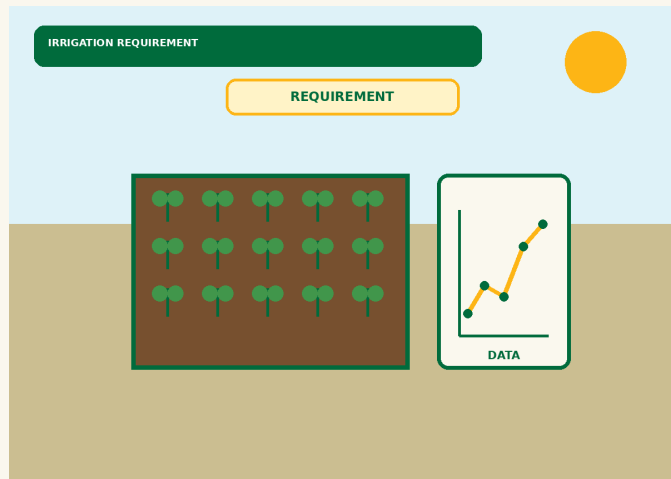
CONCEPT

Crop water use changes with growth stage, canopy, weather, rooting, and management.

INVESTIGATION TASK

Compare expected demand at two growth stages.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



REQUIREMENT

IRRIGATION REQUIREMENT

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 712 • IRRIGATION REQUIREMENT

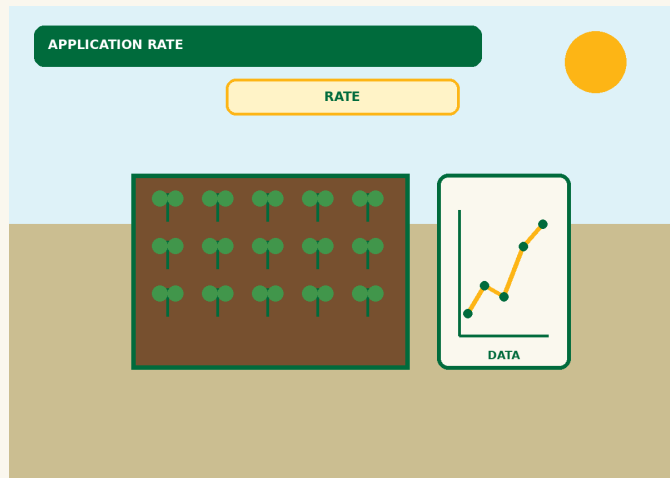
CONCEPT

Irrigation requirement is the amount of supplemental water needed after considering crop demand and effective water supply.

INVESTIGATION TASK

Solve a simplified water-balance example.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



RATE

APPLICATION RATE

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 713 • APPLICATION RATE

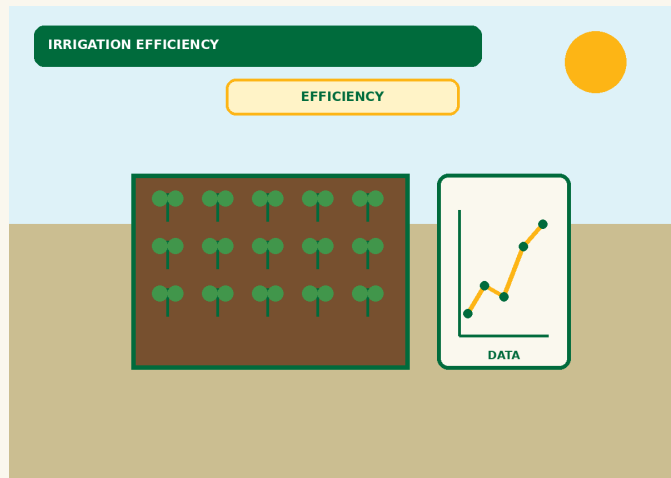
CONCEPT

Application rate describes irrigation depth or volume delivered per unit time or area.

INVESTIGATION TASK

Calculate a simple application rate.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



EFFICIENCY

IRRIGATION EFFICIENCY

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 714 • IRRIGATION EFFICIENCY

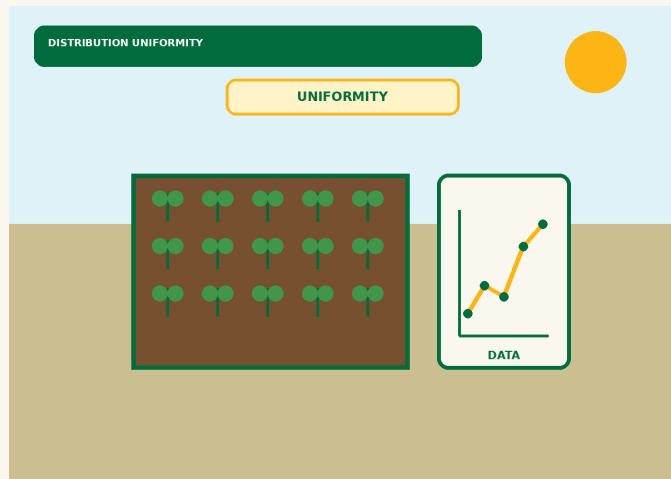
CONCEPT

Irrigation efficiency compares useful water delivery or storage with the water supplied, using a clearly defined basis.

INVESTIGATION TASK

Calculate a simplified efficiency value.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



UNIFORMITY

DISTRIBUTION UNIFORMITY

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 715 • DISTRIBUTION UNIFORMITY

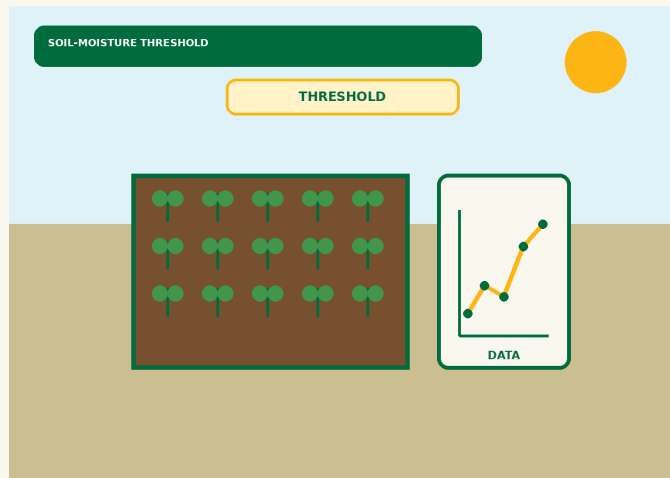
CONCEPT

Distribution uniformity describes how evenly an irrigation system delivers water across the irrigated area.

INVESTIGATION TASK

Compare emitter catch volumes.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



THRESHOLD

SOIL-MOISTURE THRESHOLD

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 716 • SOIL-MOISTURE THRESHOLD

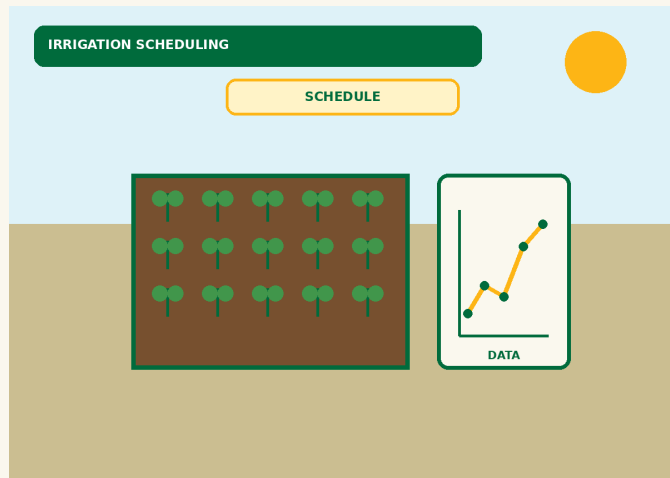
CONCEPT

A threshold is a defined soil-moisture condition used to trigger or delay irrigation.

INVESTIGATION TASK

Apply a threshold rule to monitoring data.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



SCHEDULE

IRRIGATION SCHEDULING

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 717 • IRRIGATION SCHEDULING

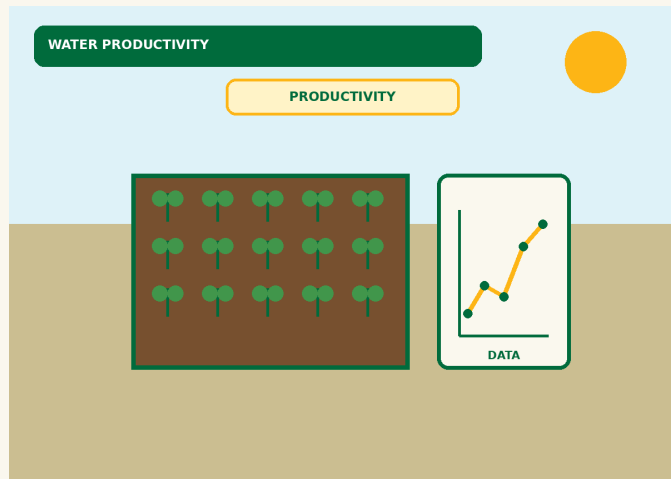
CONCEPT

Scheduling determines when irrigation occurs and how much water is applied using crop, soil, weather, and system information.

INVESTIGATION TASK

Develop a simple schedule from provided data.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



PRODUCTIVITY

WATER PRODUCTIVITY

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 718 • WATER PRODUCTIVITY

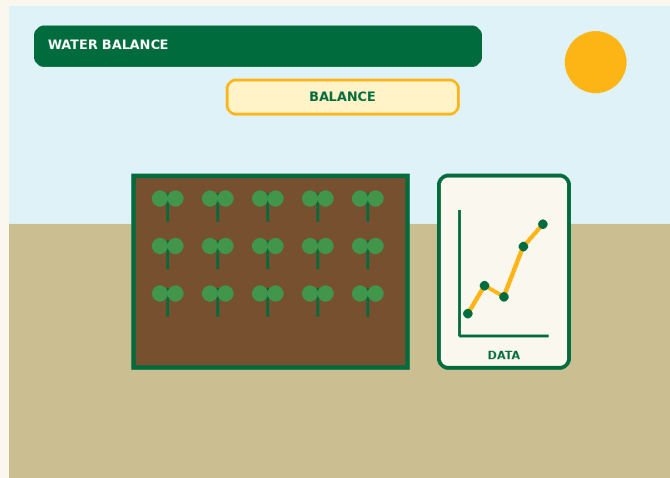
CONCEPT

Water productivity relates useful crop output to the quantity of water consumed or supplied, depending on the defined metric.

INVESTIGATION TASK

Calculate yield per unit irrigation water.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



BALANCE

WATER BALANCE

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 719 • WATER BALANCE

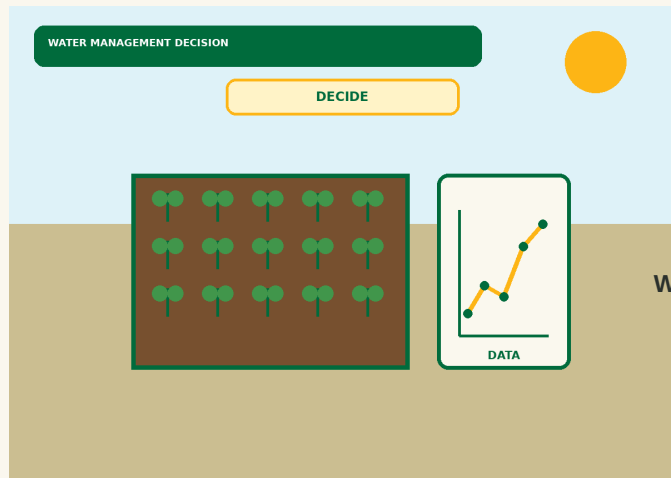
CONCEPT

A water balance accounts for inputs, outputs, and changes in stored water within a defined system boundary.

INVESTIGATION TASK

Complete a simplified root-zone balance.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



DECIDE

WATER MANAGEMENT DECISION

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 720 • WATER MANAGEMENT DECISION

CONCEPT

Water management decisions integrate monitoring, crop need, system performance, constraints, and production objectives.

INVESTIGATION TASK

Defend an irrigation decision using measured evidence.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE